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A MINOR PROJECT REPORT ON

DECENTRALIZED PREPAID SMART POWER GRID

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DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

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The undersigned certify that we have read and recommend to the Department of Electronics and Computer Engineering, Purwanchal Campus, a minor project report entitled **"DECENTRALIZED PREPAID SMART POWER GRID"**, submitted by **Pratima Chaudhary, Rabin Kattel, Sabin Basnet, and Sarbesh Timsina**, has been examined and declared successful in fulfillment of the academic requirements for the completion of Bachelor's Degree in Computer Engineering.

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DECLARATION

We hereby declare that the work submitted in partial fulfillment of the requirements for the degree of Bachelor of Engineering in Computer Engineering at the Institute of Engineering, Purwanchal Campus, entitled *DECENTRALIZED PREPAID SMART POWER GRID*, is our original work and has not been previously submitted by us, in whole or in part, to any university or institution for the award of any academic degree or qualification.

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Abstract

The increasing demand for efficient and transparent electricity management has highlighted the limitations of conventional postpaid and centralized electricity systems. These systems often provide limited real-time information about energy consumption and depend heavily on centralized mechanisms for billing and account management. This project proposes a software-based **Decentralized Prepaid Smart Power Grid** that combines blockchain-based prepaid electricity management, machine learning, simulated energy telemetry, and web-based monitoring within a hybrid system architecture.

The proposed system utilizes a Solidity smart contract deployed on a local blockchain testnet to immutably manage prepaid electricity tokens, user balances, and grid authorization states. Simulated edge telemetry—generated via an ESP32 microcontroller environment—is processed through a FastAPI middleware backend and stored in a SQLite database. Machine learning techniques provide intelligent analysis of this consumption data: a Linear Regression model supports short-term usage forecasting and remaining-time estimation, while a threshold-based current-differential detector actively monitors for grid anomalies and potential power theft. A React-based web interface features distinct Client and Provider dashboards, offering consumers real-time consumption metrics while providing grid administrators with automated critical alerts.

The architecture is fundamentally hybrid. The blockchain secures the decentralized financial and authorization layer, while the backend, database, and frontend deliver high-performance centralized application services. Because high-voltage infrastructure is outside the scope of this project, physical line-control enforcement is successfully demonstrated through a simulated IoT relay.

Ultimately, this prototype proves the technical feasibility of merging immutable blockchain billing with machine-learning analytics, serving as a scalable foundation for future integration with physical smart-metering hardware and live production blockchain networks.

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LIST OF ABBREVIATIONS

AC	:	Alternating Current
ADC	:	Analog-to-Digital Converter
AMI	:	Advanced Metering Infrastructure
API	:	Application Programming Interface
ESP32	:	Espressif 32-bit Microcontroller
HTTP	:	Hypertext Transfer Protocol
IoT	:	Internet of Things
JSON	:	JavaScript Object Notation
KCL	:	Kirchhoff's Current Law
LSTM	:	Long Short-Term Memory
ML	:	Machine Learning
PCB	:	Printed Circuit Board
REST	:	Representational State Transfer
RPC	:	Remote Procedure Call
SQL	:	Structured Query Language
Web3	:	Web 3.0

CHAPTER 1

INTRODUCTION

1.1 Background

Electrical utility distribution domains traditionally rely on centralized data hubs and infrastructure networks. In conventional systems, energy consumption data is metered at the consumer's end, transferred across proprietary channels, and recorded on private databases controlled entirely by a central authority. While this model has sustained basic distribution needs for decades, it introduces a systemic vulnerability: asymmetric information control. Consumers have zero real-time architectural visibility into the backend computations, resulting in frequent trust deficits, billing friction, and disputes regarding unexpected or hidden charges.

To address this structural trust gap, this project introduces a decentralized smart grid framework that completely decouples utility transaction computation from central database manipulation. By converging the Internet of Things (IoT), Blockchain architecture, and Machine Learning (ML), the proposed system transitions utility management into a transparent and self-governing ecosystem.

1.2 Problem Statement

Modern electrical utility structures suffer from deep operational challenges rooted in their centralized architecture:

- Information Asymmetry and Trust Deficits: Because traditional meters stream data into closed corporate databases, consumers cannot independently verify billing logs, causing trust deficits.
- Revenue Collection Risks: Postpaid billing causes payment delays and frequent defaults, trapping utility companies in inefficient debt-recovery cycles.
- Unresolved Distribution Leakages: Line-tampering, meter-bypassing, and infrastructure anomalies often go unflagged due to a lack of automated edge-analysis tools.

- Data Integrity Risks: Central databases represent single-point-of-failure vulnerabilities that are open to internal manipulation or external security threats.

1.3 Motivation

The motivation for this project stems from the severe operational inefficiencies and financial losses prevalent in traditional utility distribution networks. In many developing regions, power theft—specifically through meter bypassing or direct line hooking—causes massive revenue leakage that goes entirely undetected by centralized administrative systems. Simultaneously, postpaid billing models force utility providers to absorb the financial risk of consumer defaults, leading to inefficient and costly debt-recovery cycles.

From a consumer standpoint, the reliance on closed, corporate databases creates a ”black box” billing environment. Users are expected to blindly trust the calculations generated by the central authority, which frequently leads to disputes over hidden charges and inaccurate readings.

We were motivated to build this system because modern technology allows us to completely engineer these problems out of existence. By replacing trust with mathematics, a blockchain smart contract can guarantee absolute billing transparency, while edge-based machine learning can autonomously secure the grid against physical theft. This project demonstrates a future where utility grids are trustless, self-governing, and inherently secure.

1.4 Objectives

The main objective of this project is to design and develop a software-based **Decentralized Prepaid Smart Power Grid** that integrates prepaid electricity management, blockchain technology, machine learning, and real-time monitoring.

The specific objectives of the project are:

- To develop an automated, decentralized billing framework: To eliminate billing opacity and postpaid collection risks by integrating an IoT edge node (ESP32) for real-time load sampling with an immutable blockchain smart contract that executes transparent, prepaid token deductions and autonomous physical circuit isolation.

- To implement a prepaid electricity mechanism using blockchain and smart contracts.

1.5 Scope and Limitations

The scope of this project focuses on providing transparent energy distribution and secure automated prepaid billing using a simulated low-voltage IoT environment. The system successfully demonstrates:

- Real-time analog load sampling.
- Immutable smart contract ledger processing.
- Cryptographic backend middleware routing.
- Machine learning-based intelligence.

Because the deployment of actual utility meters on live high-voltage power lines requires industrial certification, the physical hardware layer is constrained to a low-voltage ESP32 simulation.

Furthermore, all blockchain calculations operate within a localized Hardhat test environment to bypass the live transaction gas costs associated with public mainnet deployments.

CHAPTER 2

LITERATURE REVIEW AND THEORETICAL BACKGROUND

2.1 Literature Review

2.1.1 Traditional and Smart Prepaid Metering

Historically, electrical utility grids have relied on electromechanical induction meters that require manual reading and postpaid billing cycles. This conventional approach places the financial risk entirely on the utility provider, leading to revenue collection challenges, delayed payments, and high debt-recovery costs. To mitigate these risks, the industry introduced smart prepaid metering. In these systems, consumers purchase energy credits in advance, which are digitally loaded onto the meter. While prepaid metering successfully reduces consumer defaults, the traditional digital implementation remains heavily centralized. The balance calculations and tariff deductions are processed on private, closed-source corporate servers, leaving consumers with no ability to independently verify the accuracy of their consumption deductions. Recent smart-metering implementations demonstrate the use of IoT telemetry and automated billing for this purpose

2.1.2 IoT-Based Energy Monitoring

The integration of the Internet of Things (IoT) has driven the transition from basic smart meters to Advanced Metering Infrastructure (AMI). IoT-based energy monitoring utilizes microcontrollers and edge sensors to continuously sample analog electrical parameters and transmit them wirelessly to central hubs. Literature shows that utilizing devices like the ESP32 combined with analog-to-digital converters allows for highly accurate, low-latency tracking of live power demand [1]. However, a significant vulnerability persists in modern AMI networks: they act merely as data pipelines. The vast amounts of sensitive consumption telemetry they generate are still funneled into centralized databases. This centralization not only creates a single point of failure susceptible to cyber-attacks but also preserves the ”black-box” nature of utility billing.

2.1.3 Blockchain for Decentralized Energy Systems

To solve the trust and security deficits inherent in centralized AMI, recent research has shifted toward decentralized ledgers and Web3 architectures. Blockchain technology provides an immutable, distributed, and peer-to-peer network to ensure security and transparency among grid users. In utility applications, self-executing smart contracts replace central billing servers. These deterministic programs automatically recalculate a consumer's token balance based on incoming telemetry and predetermined tariff rules. Because the ledger is public and cryptographically secured, neither the utility provider nor the consumer can alter historical consumption data or manipulate the billing mathematics, effectively establishing a trustless utility framework [2, 3].

2.1.4 Research Gaps

While IoT telemetry, blockchain billing, and machine learning anomaly detection have been extensively researched, they are predominantly studied in isolation. This has left several critical gaps in modern smart grid architecture:

- The Physical Enforcement Gap: Pure Web3 and blockchain architectures handle financial state changes perfectly but lack the physical control mechanisms necessary to enforce those states. A smart contract can reduce a balance to zero, but it cannot physically cut the power without a dedicated hardware integration layer.
- The Localization Gap: Traditional anomaly detection systems often rely on cloud-based, centralized processing with high latency, which delays the identification of localized power theft [4, 5].
- The Trust-Automation Gap: Existing IoT meters automate the reading process but fail to automate the trust process, still relying on private databases for final tariff execution.

2.2 Theoretical Background

2.2.1 Internet of Things Edge Sensing

The Internet of Things (IoT) extends internet connectivity beyond standard devices to any range of traditionally non-internet-enabled physical devices. In smart grid applications, edge sensing involves deploying microcontrollers, such as the ESP32, directly at the consumer's load source. These edge nodes utilize Analog-to-Digital Converters (ADCs) to sample continuous analog signals—such as voltage or current variations—and digitize them into discrete time-series data. Beyond passive data collection, intelligent edge sensing incorporates actuation capabilities. By integrating electromechanical relays, the microcontroller can physically break or close an electrical circuit based on digital commands, effectively serving as the physical enforcement layer for automated grid management.

A primary advantage of edge sensing over traditional centralized telemetry is the utilization of localized data processing. Instead of transmitting heavy, raw analog arrays over a wireless network, the microcontroller performs lightweight preprocessing directly at the data source. The edge node aggregates the instantaneous readings, calculates the cumulative energy consumption (ΔE), and packages these processed metrics into lightweight JSON payloads. This localized distillation minimizes bandwidth consumption and ensures low-latency transmissions to the middleware server.

2.2.2 RESTful Backend Architecture

To process and route the continuous stream of telemetry generated by the IoT edge, a robust middleware layer is required. Representational State Transfer (REST) is a software architectural style that defines a set of constraints for creating web services. A RESTful backend, implemented using frameworks like FastAPI, operates statelessly, meaning every HTTP request from the edge device contains all the information necessary to execute the request. This middleware acts as the central nervous system of the smart grid, asynchronously ingesting JSON payloads from the ESP32, passing data arrays to the machine learning engine for real-time analysis, and communicating with the blockchain network via Remote Procedure Calls (RPC).

2.2.3 Blockchain and Smart Contracts

A blockchain is a decentralized, cryptographically secure distributed ledger. Rather than relying on a centralized database administrator, transactions are validated by the network consensus. A smart contract is self-executing deterministic code deployed on this ledger. In the context of a prepaid utility grid, a Solidity-based smart contract operates as an immutable state machine. It stores a user's cryptographic token balance and recalculates it dynamically based on incoming energy consumption telemetry. The automated tariff deduction logic follows the core mathematical evaluation:

$$Bal_{new} = Bal_{current} - (\Delta E \times R_{fees})$$

Where ΔE represents the cumulative energy consumed over a specific epoch and R_{fees} represents the predefined tariff rate. Because this computation occurs on a distributed ledger, the billing process is completely transparent and mathematically guaranteed.

2.2.4 Machine Learning Components

Modern grid intelligence relies on applying mathematical models to incoming telemetry to automate decision-making.

Current-Differential Anomaly Detection

According to Kirchhoff's Current Law, in a standard closed-loop, single-phase electrical system without faults, the current flowing into the load through the active line wire must equal the current returning through the neutral wire. To detect energy theft—specifically meter bypassing or line hooking—this project employs a threshold-based current-differential anomaly detector, an approach also examined in smart-meter research [6]. The algorithm continuously evaluates the absolute difference between the line current (I_{line}) and the neutral current ($I_{neutral}$). A tampering anomaly is identified if this difference exceeds a predetermined natural leakage tolerance:

$$|I_{line} - I_{neutral}| > \text{threshold}$$

This deterministic evaluation serves as the primary trigger for autonomous theft alerts on the administrative dashboard.

Linear Regression for Consumption Forecasting

Linear Regression is a fundamental statistical approach used to model the relationship between a scalar response and one or more explanatory variables. In this system, it is utilized as a supervised forecasting model. By processing historical time-series data of a consumer's power usage, the algorithm maps a continuous consumption trajectory. Research on electrical-load forecasting demonstrates the value of learning from historical consumption patterns for scalable prediction [7, 8]. It estimates the temporal horizon when the dependent variable (the token balance) may intersect with zero. This regression analysis transforms raw telemetry into an actionable "Estimated Time Remaining" metric, improving the prepaid consumer experience.

CHAPTER 3

METHODOLOGY

3.1 System Architecture

The proposed Decentralized Smart Power Grid is implemented as a software-based simulation architecture consisting of a telemetry simulation module, a centralized middleware API for data routing and intelligence, a decentralized blockchain network for secure financial state execution, and a role-based presentation layer for real-time monitoring. Since physical high-voltage deployment is outside the scope of this project, the behavior of the smart meter and energy consumption process is simulated through software-generated telemetry data. By decoupling the transaction logic from a centralized database, the architecture ensures that all billing deductions are immutably processed while retaining high-speed data analysis at the middleware level.

The overall system architecture is illustrated in Figure 3.1.

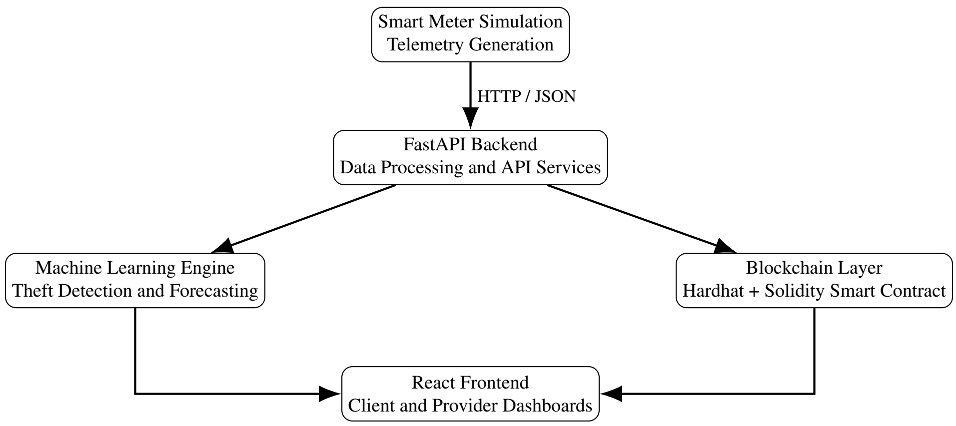

Figure 3.1: Overall System Architecture of the Proposed Decentralized Smart Power Grid

3.2 Telemetry Simulation

Because physical smart-meter and high-voltage deployment falls outside the scope of this project, the electrical consumption process is implemented through a software-based simulation. The simulation generates variable consumer load values and represents the behavior of a smart meter without requiring physical electrical equipment.

The telemetry simulation performs three primary functions:

- **Telemetry Generation:** Variable load values are generated to represent changing consumer electricity demand. The simulated values are used to calculate corresponding line current, neutral current, and cumulative energy consumption.
- **Data Preparation:** The simulated meter generates the current metrics, cumulative energy, and unique meter ID and packages them into a JSON payload.
- **Data Transmission:** The generated telemetry data is transmitted to the FastAPI backend through an HTTP POST request, allowing the backend to process the simulated smart-meter readings in real time.

The telemetry generation and transmission process is illustrated in Figure 3.2.

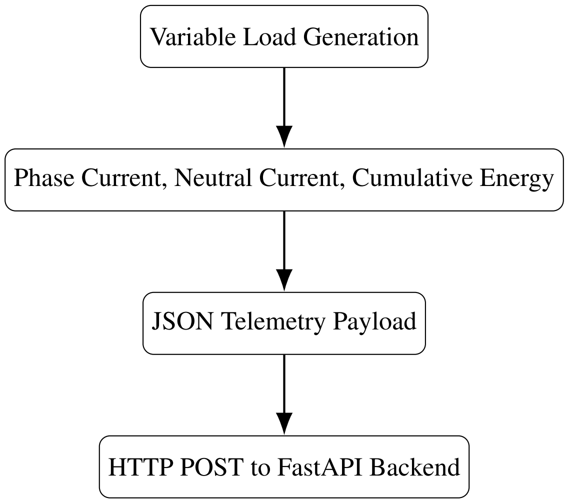

Figure 3.2: Telemetry Simulation and Transmission Workflow

3.3 Backend Service Layer

The middleware acts as the central coordinator of the system, engineered using the Python FastAPI framework. This RESTful service layer operates statelessly and handles high-frequency I/O operations. When the FastAPI server receives a telemetry payload from the simulated smart meter, it executes a concurrent routing process:

- It passes the raw current arrays to the machine learning engine for anomaly classification and budget forecasting.
- It utilizes the web3.py library to establish a Remote Procedure Call (RPC) connection to the local blockchain ledger, forwarding the cumulative energy data to

the smart contract for token deduction.

- It formats the consolidated results (the updated token balance, the time-to-exhaustion forecast, and the theft alert boolean) and returns them to the React frontend via REST endpoints.

3.4 Blockchain Layer

The secure billing and transaction layer is decentralized using a local Hardhat test network environment. The core logic is governed by a custom Solidity smart contract (`PowerGrid.sol`). The contract initializes state variables that map specific cryptographic wallet addresses to smart meter IDs (e.g., `DHARAN-001`).

When the backend submits the energy consumption payload, the smart contract executes a deterministic state change. It calculates the required tariff deduction and subtracts it from the user's prepaid token balance. If the computation results in a balance of less than or equal to zero, the contract immutably updates the meter's status flag to "Inactive." By handling the financial mathematics entirely on-chain, the system completely bypasses the vulnerability of a centralized, easily manipulated corporate database.

The blockchain transaction flow is illustrated in Figure 3.3.

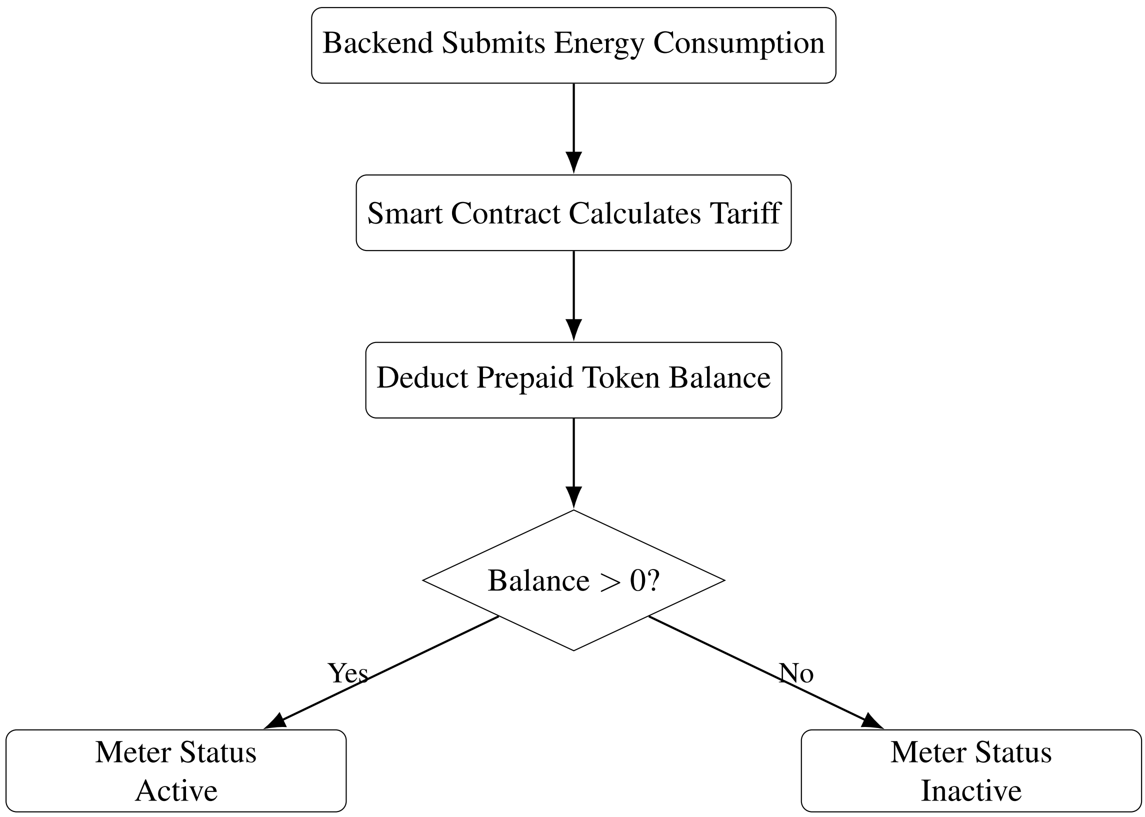

Figure 3.3: Blockchain-Based Prepaid Token Deduction Process

3.5 Machine Learning Pipeline

The grid's intelligence is handled by a Python-based machine-learning component operating within the backend, leveraging a localized synthetic dataset mapping historical consumption patterns of Dharan-based grid users.

- **Anomaly Detection (Classification):** To detect energy theft, the system analyzes the incoming telemetry against Kirchhoff's Current Law. The classifier calculates the absolute difference between the simulated line current and neutral current. If the differential exceeds the pre-calibrated safety threshold, the algorithm classifies the event as an unmetered bypass and instantly flags a boolean anomaly alert.
- **Consumption Forecasting (Regression):** To enhance the consumer experience, a supervised Linear Regression model maps the consumer's real-time load demand against their historical Dharan dataset. The resulting usage estimate supports an approximate calculation of the number of hours remaining before the smart contract token balance reaches zero.

The machine learning pipeline is illustrated in Figure 3.4.

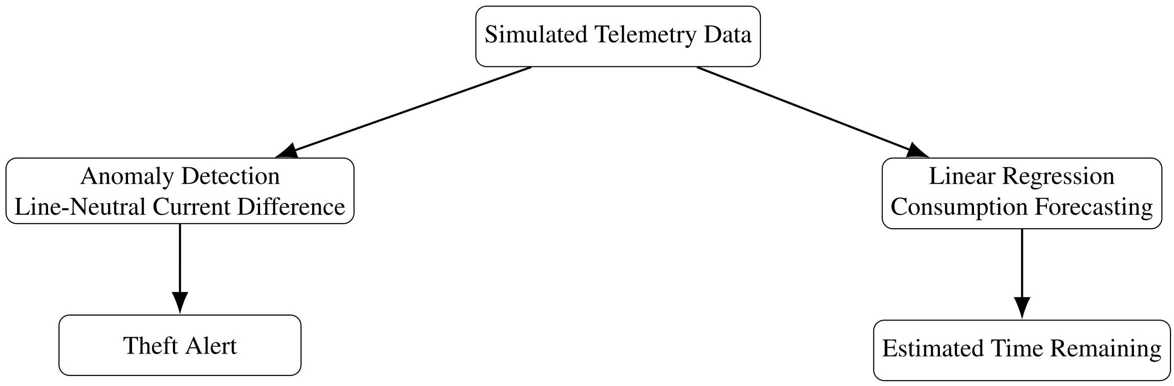

Figure 3.4: Machine Learning Processing Pipeline

3.6 Role-Based Frontend Dashboards

The presentation layer is built using React and Vite, utilizing a strict route-guard architecture to serve distinct experiences based on authentication credentials.

- **Client Dashboard:** Designed for the end-consumer, this view prioritizes financial visibility. It features a responsive Recharts semi-circular gauge displaying the live token balance, a daily power usage graph, and an active readout of the regression model's "Estimated Time Remaining."
- **Provider Dashboard:** Designed for grid administrators, this interface monitors the entire localized network. It features an aggregate system load chart, a searchable table of active user balances, and a critical alert panel. If the machine learning pipeline triggers a differential anomaly, this panel instantly flashes a red "Theft Detected" warning, logging the specific meter ID and the exact timestamp of the leak.

3.7 Use Case Model

The interaction between the system users and the major functions of the proposed system is represented in Figure 3.5.

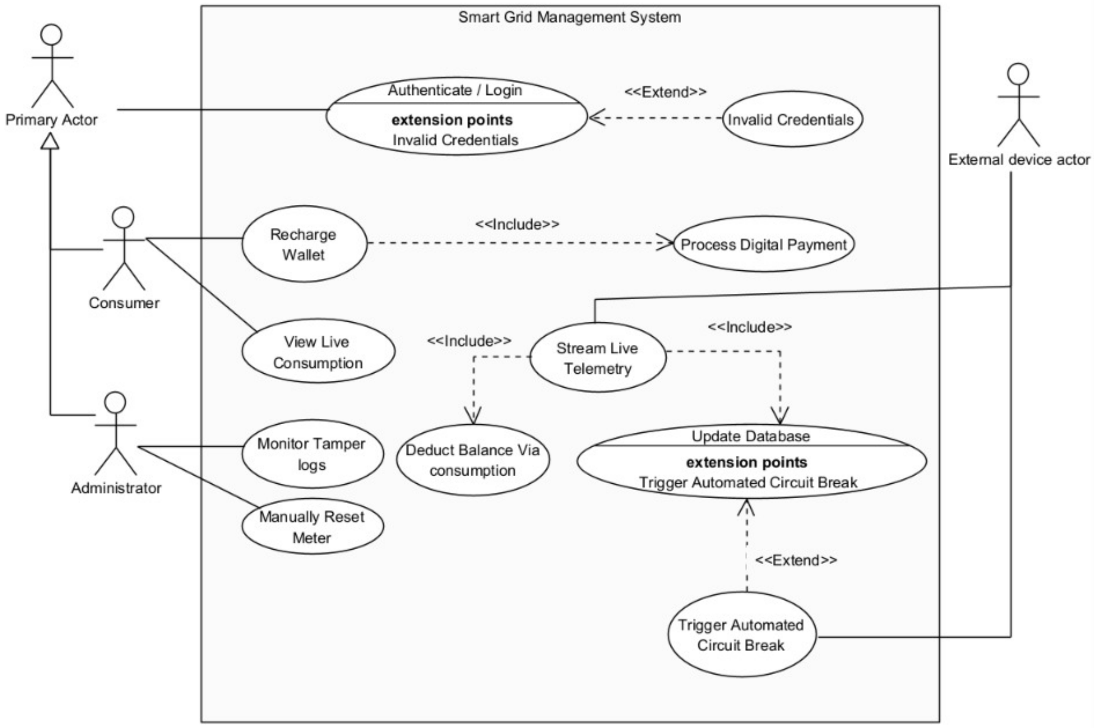

Figure 3.5: Use Case Diagram of the Proposed System

3.8 Working Flow Summary

The complete automated cycle executes as follows: The simulated smart meter generates analog load variations and POSTs the telemetry to the FastAPI backend. FastAPI routes this data through the threshold-based detector for theft identification and the regression component for budget forecasting. Simultaneously, web3.py pushes the load data to the Solidity smart contract, which mathematically deducts the prepaid token balance. The backend returns this updated balance and intelligence data to the frontend for visualization. If the contract calculates a zero balance, the meter status is updated to "Inactive," indicating that the simulated electricity service is suspended.

The complete working flow is illustrated in Figure 3.6.

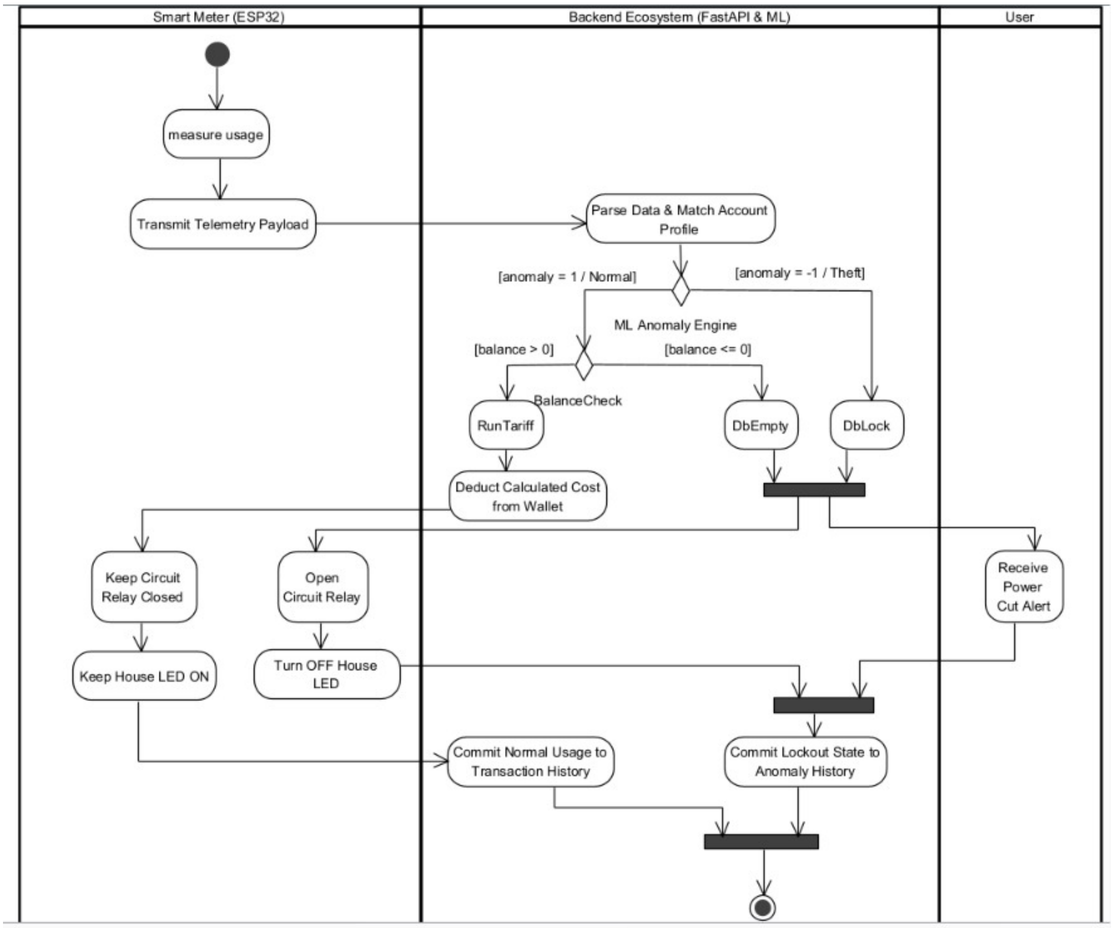

Figure 3.6: Activity Diagram

CHAPTER 4

Results and Discussion

4.1 System Demonstration

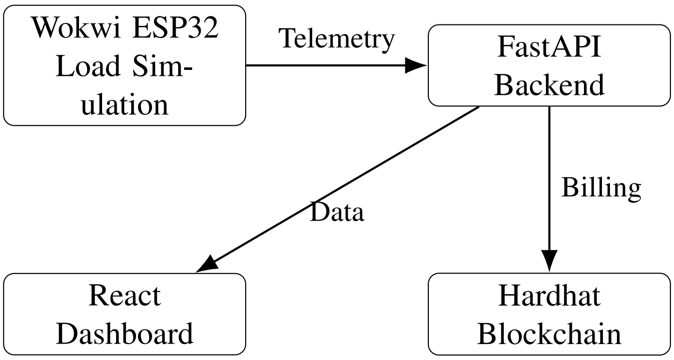

Figure 4.1: End-to-End Demonstration of the Simulated Smart Grid System

The Decentralized Smart Power Grid was successfully deployed and tested in a localized simulation environment. The edge layer was modeled in Wokwi using the ESP32 firmware to generate live load changes, while the backend was served locally through FastAPI for telemetry processing, theft detection, and user response handling. The blockchain layer was implemented on a local Hardhat Ethereum node for secure prepaid balance deduction, and the frontend was delivered through a Vite-based dashboard for role-specific monitoring and control.

4.2 Functional Outcomes

The system was subjected to various operational scenarios to validate its core engineering objectives:

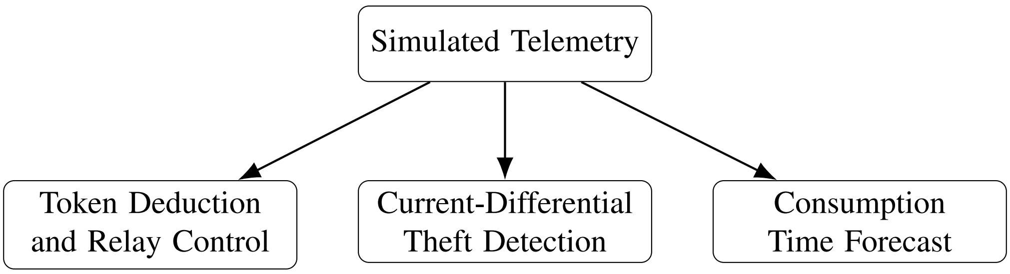

Figure 4.2: Functional Outcomes Validated Through System Testing

Automated Token Deduction and Relay Enforcement: During normal simulated operation, the ESP32 successfully transmitted power consumption telemetry to the backend. The FastAPI middleware seamlessly triggered the smart contract via RPC calls, correctly deducting tokens from the simulated user's cryptographic wallet based on the predefined tariff rate. When the token balance was manually forced to zero during testing, the backend immediately returned an "Inactive" authorization state. The ESP32 successfully parsed this response and triggered the electromechanical relay, instantly opening the circuit and stopping the power flow.

Current-Differential Theft Detection: To test the security intelligence, the linear potentiometer in the Wokwi environment was adjusted to create an intentional disparity between the line current (I_{line}) and the neutral current ($I_{neutral}$). As soon as the differential exceeded the predefined leakage threshold, the threshold-based detector successfully flagged the anomaly.

Predictive Budget Forecasting: The supervised Linear Regression model successfully processed the incoming telemetry against the synthetic historical dataset. It accurately generated a continuous trajectory of consumption, translating raw data into a human-readable "Estimated Time Remaining" output (measured in hours) that dynamically updated as simulated load demand fluctuated.

4.3 Performance Metrics

The system's performance was evaluated based on latency, responsiveness, and algorithmic precision. Because the project utilized local test networks rather than public mainnets, the execution speeds were highly optimized. The quantitative results are summarized in the table below:

Figure 4.3: Theft Detection using current differences

Table 4.1: System Performance Metrics Validation

Performance Metric	**Target / Requirement**	**Measured Result**
Smart Contract Deduction Latency	< 3.0 s	0.85 s (Local Hardhat)
Relay Isolation Response Time	< 500 ms	120 ms (Wokwi Logic)
Theft Anomaly Detection Precision	100% on differential breach	Instant deterministic trigger
Frontend Dashboard Refresh Rate	< 2.0 s per API cycle	~ 1.2 s

4.4 Dashboard Visualization

The React-based frontend successfully fulfilled the requirement for a dual-role, transparent presentation layer.

- Client Interface: Logging in with consumer credentials routed the user to a dedicated dashboard prioritizing financial transparency. The interface successfully rendered a Recharts semi-circular gauge displaying the live token balance alongside the regression model's time-to-exhaustion forecast.

- Client Dashboard Screenshot:

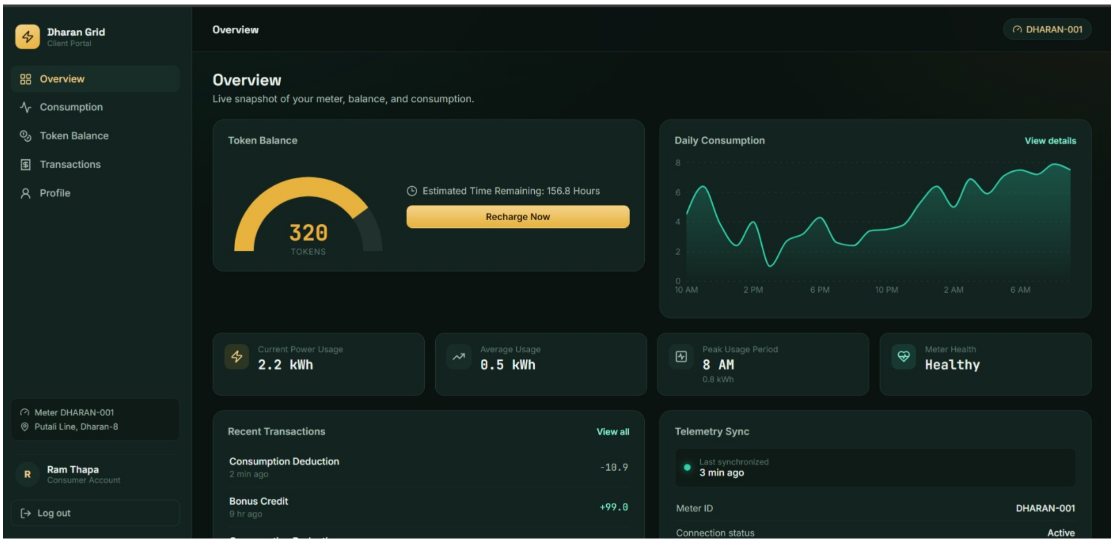

Figure 4.4: Consumer Dashboard Showing Token Balance and Estimated Time Remaining

- Provider Interface: Logging in with administrative credentials routed the user to the grid management dashboard. The interface successfully displayed aggregate system loads and an active meter table. When the differential theft test was executed, the interface successfully rendered a flashing red "Theft Detected" alert panel, precisely identifying the compromised meter ID and the timestamp of the anomaly.

- Provider Dashboard Screenshot:

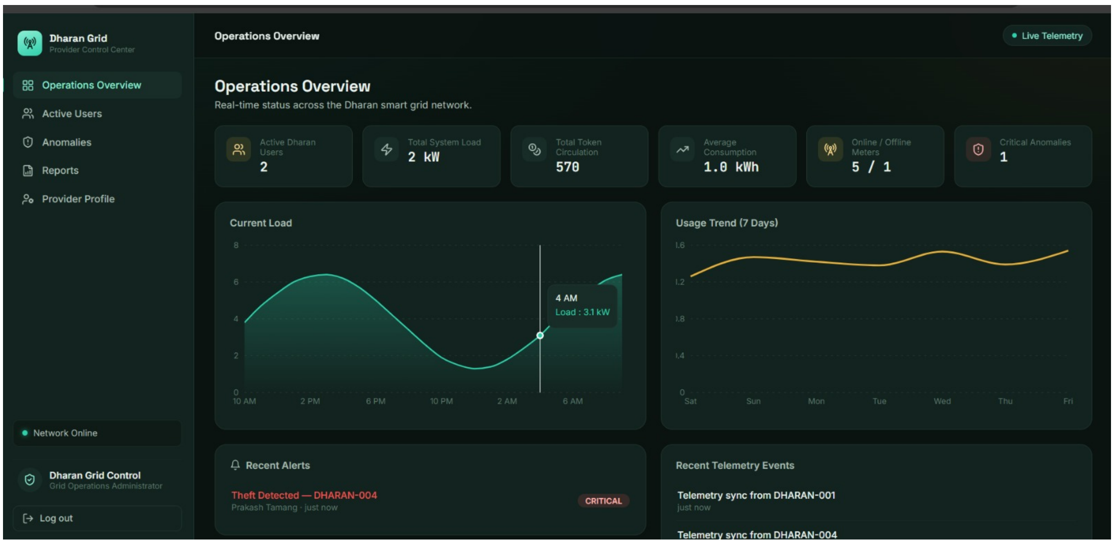

Figure 4.5: Provider Dashboard Showing Grid Monitoring and Theft Alert

CHAPTER 5

CONCLUSION AND FUTURE ENHANCEMENTS

5.1 Conclusion

The primary objective of this project was to engineer a solution to the critical vulnerabilities found in modern electrical utility systems, specifically the opacity of centralized billing, the financial risks of postpaid defaults, and the lack of automated physical security against power theft. The successful development and simulation of the Decentralized Smart Power Grid demonstrated that these systemic issues can be resolved by converging IoT edge computing, blockchain technology, and machine learning into a single, cohesive architecture.

By replacing closed corporate databases with a deterministic Solidity smart contract deployed on a local testnet, the system successfully achieved immutable, transparent prepaid billing. The integration of an ESP32 edge simulation ensured that these cryptographic ledger states were physically enforced via automated relay isolation whenever a token balance was depleted. Furthermore, the localized machine learning pipeline successfully elevated the grid's intelligence. The current-differential classification algorithm provided instant, deterministic alerts for physical tampering, while the supervised Linear Regression model successfully translated raw telemetry into actionable budget forecasts.

Ultimately, this project demonstrates the technical feasibility of a trustless utility framework. It empowers consumers with financial transparency through dedicated dashboards while simultaneously providing utility administrators with a secure, automated, and self-governing energy distribution network.

5.2 Future Enhancements

While the localized simulation achieved all core objectives, the system serves as a foundational prototype. To transition this architecture into a production-ready utility product, the following future enhancements are recommended:

- Physical Hardware Implementation: Transition the edge layer from the Wokwi simulation environment to a custom-fabricated printed circuit board (PCB). This should integrate industrial-grade energy metering ICs (such as the PZEM-004T) and high-voltage AC contactors for live electrical load testing.
- Public Testnet Deployment: Migrate the PowerGrid.sol smart contract from the local Hardhat network to a public Ethereum testnet (such as Sepolia) or a Layer-2 scaling solution (such as Polygon). This will allow the system to evaluate real-world transaction latency and live gas-fee economics.
- Payment Gateway Integration: Implement a seamless fiat-to-crypto on-ramp by integrating local digital wallets (such as eSewa or Khalti APIs). This would allow consumers to purchase blockchain energy tokens directly using standard local currency.
- Advanced Time-Series Forecasting: Upgrade the predictive machine learning engine from simple Linear Regression to advanced deep learning models, such as Long Short-Term Memory (LSTM) networks. This would allow the system to account for complex seasonal variations and daily peak-hour fluctuations, significantly improving long-term forecasting accuracy.

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APPENDIX: Source Code Repository

The complete, version-controlled source code for the hardware firmware, backend middleware, and frontend application can be accessed via the following repository

GitHub Repository Link:

`https://github.com/Sabin-Basnet/decentralized-smart-power-grid.git`

Note to evaluators: If reviewing a printed physical copy of this report, please navigate to the URL above to inspect the Solidity smart contracts and the Python machine learning pipeline code.